

Computational Bootcamp Syllabus

John P. Ryan

Summer 2026

Instructor: John P. Ryan, john.p.ryan@wisc.edu

Instructional Mode: Hybrid in-person and virtual

Lectures: Monday/Wednesday, 1–2:15pm (location TBD)

Office Hours: Monday/Wednesday, 2:30–3:30pm (Sewell 7143)

Course Description

This four-week course quickly familiarizes students with computational methods and emphasizes their applications to Economics. By the end of the course, students will learn what computation is, how it applies to Economics, and how to do it. The course will be taught in the Julia programming language. Rising second year PhD students are the target audience, but all are welcome (and encouraged!) to attend.

This course meets for two 75-minute lectures each week (date TBD). Lectures will be held on Monday and Wednesday from 1–2:15pm. The course carries the expectation that students will work on course learning activities for about 6 hours every week outside of class.

Outcomes

- Learn computational methods for structural modelling and estimation in economics
- Gain familiarity with dynamic programming and solution methods
- Understand best practices for programming, performance, and version control
- Obtain proficiency in the Julia programming language
- Introduce students to resources for computation available at UW-Madison, especially the SSCC

Homeworks and “Grading”

Problem sets will be weekly and will require using Julia to execute a variety of tasks. The “grade” for the class is entirely based on the four problem sets, each which will be given equal weight.

This course is entirely optional and is solely intended to help you learn how to develop your computational skills. The homeworks are a way of guiding your development, but are not meant to cause additional stress.

None of the “grades” in this course will go on your transcript, none of your professors will see your grades, and you are free to choose whatever level of participation in this course that you like. The resources are all uploaded to the course website so that you can access them at your convenience.

AI Policy

For the first three problem sets, I ask that you *not* use AI assistants (Copilot, ChatGPT, Claude, and the like) to write your solutions. The goal of this bootcamp is to build fluency in writing and debugging code yourself, and the early weeks are a clean, low-pressure environment in which to do that — struggling through the problems is where the learning happens. Because the course is not for credit, this is a request rather than an enforced rule.

In Week 4, we discuss AI tools directly: Lecture 7 covers where they help and where they fail, and Homework 4 includes an exercise in using one deliberately and verifying its output.

Additional Resources

- Documentation for the Julia language is at <https://docs.julialang.org/en/v1/>.
- [Julia performance tips](#) from the documentation. Can be very helpful!
- Material in this course borrows considerably from <https://julia.quantecon.org/>. Additional helpful resources and exercises may be found there.
- [Jesus Fernandez-Villaverde's lecture notes](#) on computation (for Julia and many other things)
- [Dean Corbae's Computational Economics course](#), that this course should prepare you for

Schedule

Week 1: Introduction to Computation and Julia

- Lecture 1: Intro to Computation, Basic Julia, and Version Control
- Lecture 2: Performance Optimization and Advanced Julia
- Homework 1: Due Wednesday, August 5

Week 2: Numerical Methods

- Lecture 3: Numerical Optimization, Root Finding, and Fixed Points
- Lecture 4: Interpolation, Integration
- Homework 2: Due Wednesday, August 12

Week 3: Parallelization and Dynamic Programming

- Lecture 5: Parallelization and UW Resources
- Lecture 6: Dynamic Programming, Value Function Iteration
- Homework 3: Due Wednesday, August 19

Week 4: Simulation and Applications

- Lecture 7: Simulation, AI Tools and Best Practices
- Lecture 8: Economic Applications
- Homework 4: Due Wednesday, August 26

Misconduct Statement

Academic Integrity is critical to maintaining fair and knowledge-based learning at UW Madison. Academic dishonesty is a serious violation: it undermines the bonds of trust and honesty between members of our academic community, degrades the value of your degree and defrauds those who may eventually depend upon your knowledge and integrity.

Examples of academic misconduct include, but are not limited to: cheating on an examination (copying from another student's paper, referring to materials on the exam other than those explicitly permitted, continuing to work on an exam after the time has expired, turning in an exam for regrading after making changes to the exam), copying the Homework of someone else, submitting for credit work done by someone else, stealing examinations or course materials, tampering with the grade records or with another student's work, or knowingly and intentionally assisting another student in any of the above. Students are reminded that online sources, including anonymous or unattributed ones like Wikipedia, still need to be cited like any other source; and copying from any source without attribution is considered plagiarism.

The Department of Economics will deal with the offenses harshly following UWS14 procedures:

1. The penalty for misconduct in most cases will be removal from the course and a failing grade.
2. The department will inform the Dean of Students as required and additional sanctions maybe applied.
3. The department will keep an internal record of misconduct incidents. This information will be made available to teaching faculty writing recommendation letters and to admission offices of the School of Business and Engineering.

If you think you see incidents of misconduct, you should tell your instructor about them, in which case they will take appropriate action and protect your identity. You could also choose to contact our a department administrator and your identity will be kept confidential.

Grievance Procedure

The Department of Economics has developed a grievance procedure through which you may register comments or complaints about a course, an instructor, or a teaching assistant. The Department continues to provide a course evaluation each semester in every class. If you wish to make anonymous complaints to an instructor or teaching assistant, the appropriate vehicle is the course evaluation. If you have a disagreement with an instructor or a teaching assistant, we strongly encourage you to try to resolve the dispute with him or her directly. The grievance procedure is designed for situations where neither of these channels is appropriate.

If you wish to file a grievance, you should go to room 7238 Social Science and request a Course Comment Sheet. When completing the comment sheet, you will need to provide a detailed statement that describes what aspects of the course you find unsatisfactory. You will need to sign the sheet and provide your student identification number, your address, and a phone where you can be reached. The Department plans to investigate comments fully and will respond in writing to complaints.

Your name, address, phone number, and student ID number will not be revealed to the instructor or teaching assistant involved and will be treated as confidential. The Department needs this information, because it may become necessary for a commenting student to have a meeting with the department chair or a nominee to gather additional information. A name and address are necessary for providing a written response.

Accommodations for Students with Disabilities

The University of Wisconsin-Madison supports the right of all enrolled students to a full and equal educational opportunity. The Americans with Disabilities Act (ADA), Wisconsin State Statute (36.12), and UW-Madison policy (Faculty Document 1071) require that students with disabilities

be reasonably accommodated in instruction and campus life. Reasonable accommodations for students with disabilities is a shared faculty and student responsibility. Students are expected to inform faculty [me] of their need for instructional accommodations by the end of the third week of the semester, or as soon as possible after a disability has been incurred or recognized. Faculty [I], will work either directly with the student [you] or in coordination with the McBurney Center to identify and provide reasonable instructional accommodations. Disability information, including instructional accommodations as part of a student's educational record, is confidential and protected under FERPA.

Diversity and Inclusion

Diversity is a source of strength, creativity, and innovation for UW-Madison. We value the contributions of each person and respect the profound ways their identity, culture, background, experience, status, abilities, and opinion enrich the university community. We commit ourselves to the pursuit of excellence in teaching, research, outreach, and diversity as inextricably linked goals.

The University of Wisconsin-Madison fulfills its public mission by creating a welcoming and inclusive community for people from every background — people who as students, faculty, and staff serve Wisconsin and the world.

Acknowledgment

The structure and content of this course is due, in large part, to the instructors who have taught the bootcamp before me. This includes John Higgins, Kevin Hunt, and Garrett Anstreicher.